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MILITARY SPECIFICATION
RADIACMETER IM-93()/UD
RADIACMETER IM-147()/PD

This specification is approved for use by Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. Scope

1.1 Scope. This specification covers the requirements for construction and testing of the Radiacmeter IM-93()/UD and Radiacmeter IM-147()/PD.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TM, Fort Monmouth, New Jersey 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC-N/A FSC 6665

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SPECIFICATIONS

FEDERAL

L-P-590 - Plastic Molding and Extrusion Material, Polyethylene and Copolymers (Low, and High Density).

MILITARY

MIL-P-11268 - Parts, Materials, and Processes Used in Electronics Equipment
MIL-M-13231 - Marking of Electronic Items
MIL-F-14072 - Finishes for Ground Electronic Equipments

STANDARDS

MILITARY

MIL-STD-252 - Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-810 - Environmental Test Methods

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other government documents, drawings, and publication. The following government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

Handbook H75 National Bureau of Standard

DRAWINGS

SC-GL-57913 Gages for Radiacmeter IM-93()/UD
SC-GL-57914 Gages for Radiacmeter IM-147()/PD
SM-C-136052 Radiacmeter IM-93()/UD and IM-147()/PD

(Copies of specifications, standards, drawings and publications required by contractors on connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulation unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2) the contractor shall furnish first article units for first article inspection and approval (see 4.3).

3.2 Construction. The equipment shall be constructed in accordance with the requirements specified herein. The overall dimensions shall be in accordance with SM-C-136052 and the following 3.2.1 through 3.2.7 paragraphs.

3.2.1 Condenser. The dosimeter shall contain a condenser (electrical capacitor) in parallel with the electrometer and chamber wall. The condenser shall be double wrapped with two layers of 0.001 inch thick insulating film. There shall be less than one 0.001 inch diameter pinhole per inch of insulating film for each layer of 0.001 ±.0005 inch thick film; there can be no pinhole in excess of 0.0015 inches. NOTE: To test for pinholes, samples of the condenser insulating film (single layers) are examined over blotting paper which has been soaked in weak Sodium Hydroxide (NAOH) solution. Bright magenta spots on the blotter will indicate the presence or a pinhole in the capacitor film. The performance of this test is not a requirement of this specification.

3.2.2 Charging electrode diaphragm. The charging electrode diaphragm assembly shall be so constructed that the force on the charging electrode necessary to make electrical contact between it and internal electrode of the chamber shall be greater than 3 pounds but shall not exceed 6 pounds at sea level and at an ambient temperature of 70 degrees F. The electrode shall be capable of withstanding axial forces of at least 10 pounds and radial forces of at least 5 pounds (see 4.8.17).

3.2.3 Clip. The radiacmeter shall be equipped with a clip to prevent loss. The clip shall be affixed in such a manner as to prevent rotation about the barrel end to afford considerable resistance to removal therefrom. A male-female detent may be used with the male portion on the radiacmeter barrel. Any design clip may be used provided that the dosimeter may be inserted fully into the UDM-2 Calibrator and the top of the UDM-2 completely closed without removing the packet clip from the dosimeter.

3.2.4 Cap. A protective transparent cap shall be fitted on the end of the barrel to prevent accidental discharge of the radiacmeter. The cap shall be made of a polyethylene compound conforming to Type 1, Grade 2, Natural, of Federal Specification L-P-590. The material shall be sufficiently transparent to permit the reading of the dosimeter without removing the cap. It shall not exceed 5/8 inch in length, shall have thickness of approximately 1/16 inch and shall be capable of removal as required.

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3.2.5 Scale. The reticle for each Radiacmeter shall be marked by etching, engraving, a photosensitive eye, photographic process, or by other processes capable of meeting requirements of this specification. The markings for each radiacmeter scale shall be as follows:

3.2.5.1 Radiacmeter IM-93()/UD. There shall be seven heavy lines to designate the seven numbered scale positions corresponding to 0, 100, 200, 300, 400, 500 and 600 roentgens respectively. Four equally heavy shorter lines shall be used to set off the five divisions between the major numbered heavy lines. In addition there shall be an equally heavy line perpendicular to end bisecting all scale divisions. The word Roentgens and the nomenclature of the radiacmeter shall be inscribed above the scale. No other markings are permitted.

3.2.5.2 Radiacmeter IM-147()/PD. There shall be six heavy lines to designate the six numbered scale positions corresponding to 0, 10, 20, 30, 40, and 50 roentgens respectivety. Four equally heavy shorter lines shall be used to set off the five divisions between the major numbered heavy lines. In addition there shall be an equally heavy line perpendicular to end bisecting all scale divisions. The word roentgens and the nomenclature of the radiacmeter shall be inscribed above the scale., No other markings are permitted.

3.2.5.3 Materials. The scale shall be constructed of a material which will not become discolored, pitted or damaged in any way by exposure to light, heat or any ionizing radiation.

3.2.6 Optical system. Throughout the entire range, the optical system shall permit the fiber to be clearly resolved relative to the scale divisions when the radiacmeter is exposed to an illumination of 0.5 foot candles with cap (para. 3.24) removed and 0.6 foot candles with cap in place from a diffused source at the charging diaphragm. The focus of fiber and reticle shall permit resolution to 1/4 of the smallest scale division over the entire scale. (See 4.12.)

3.2.6.1 Fiber image. The image of the fiber shall appear as a vertical line to within 1/4 minor scale division, over the entire scale, i.e., with the charging potential adjusted so the images of the fiber coincides with the bottom of any major scale division marking, the distance between the image of the fiber and the top of that same major division marking shall not be greater than 1/4 the distance between two adjacent minor division markings.

3.2 7 Surface. The body of each radiacmeter shall be smooth with-out grooves or crevices except for the groove for the clip. Staking marks are acceptable providing they introduce no porosities, fissures, or breaks in the continuous anodic, finished surface of the dosimeters.

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3.3 Parts, materials, and processes; general. Parts, materials, and processes shall be such as to enable the equipment to meet its performance and other requirements, as specified herein. Ratings and derated values of parts and materials shall not be exceeded when the equipment is subjected to specified service conditions (see 3.10). Derating methods used shall be as specified by the applicable individual documents covering the parts and materials. Material capable of producing or evolving ammonia or other corrosive gases shall not be used.

3.3.1 Recycled virgin and reclaimed materials. To the maximum extent possible, recovered or reclaimed materials, in lieu of virgin, shall be used without jeopardizing the intended use of the manufactured item.

3.3.2 Seal. Each radiacmeter shall be hermetically sealed by the use of glass to metal and metal to metal vacuum (solder type) sealing techniques. The seal at the viewing window and of the instrument may be accomplished by using Hysol 4241 epoxy compound or equal, provided the final product meets full specification requirements. (See 4.8.16 and 4.9)

3.3.3 Cleaning.

3.3.3.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are assembled.

3.3.3.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.4 Finish. The equipment shall be finished with a dyed coating in accordance with MIL-F-14072 and the color shall match number 24087 of Federal Standard Number 595, if the barrel material is aluminum. A matte tin-nickel finish is acceptable if the barrel material is brass (See 4.4).

3.5 Marking.- Markings shall conform to MIL-M-13231. (See 4.4.) When marked by engraving, the indentations shall be filled flush to the surface with white non-porous paint or similar process. Filling used shall be capable of withstanding abrasion to a greater extent than the finish of the equipment.

3.5 1 Serial numbers. Each radiacmeter shall be serial numbered

3.6 Interchangeability. All parts of the same type shall be physically and functionally interchangeable. (See 4.6).

3.7 Weight. The weight of each radiacmeter shall not exceed 2 ounces (64 grams) (See 4.7).

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3.8 Safety engineering.

3.8.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.10).

3.8.2 Corner and edge rounding. All exposed corners and edges shall be rounded to eliminate possible injury to personnel due to lacerations and cuts.

3.8.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials, markings, electron tubes, surge arrestors and lenses).

3.9 Performance requirements.

3.9.1 Calibration accuracy. Each radiacmeter shall detect and measure X-rays and gamma rays to within ± 10 percent of the true dose. (See 4.8.1).

3.9.2 Linearity. The deviation from linearity using the zero and center scale readings as a datum shall not be more than 5 percent at the 1/4 and 3/4 scale levels and shall not be more than 10 percent at the full-scale level. (See 4.8.2)

3.9.3 Precision. Each radiacmeter shall be capable of reproducing its own readings to ± 2.5 percent for the IM-93()/UD and 4 percent for the IM-147()/PD when read at 30R and when subjected to a series of calibration runs at mid-scale. (See 4.8.3).

3.9.4 Energy dependence. The response of each radiacmeter shall be as uniform as possible when exposed to equal roentgen doses of X-ray and gamma radiation whose energies lie between 80 thousand electron volts (KeV) and 1.5 million electron volts (MeV) for the IM-147()/PD and between 80 KeV and 5.0 MeV for the IM-93()/UD. That is, when each radiacmeter is irradiated with equal roentgen doses of X-rays and gamma rays of energies in its specified range, and the response of the radiacmeter to each of these doses is normalized to the response of the radiacmeter to the same dose of Cobalt 60 gamma radiation, the normalized values shall all be as close to unity as possible, and shall in no case lie outside the range of 0.85 to 1.15 (±15%). For energies lower than 80 KeV, the response of each radiacmeter shall, as far as is practicable, match that of the human body, and shall in all cases exhibit a continual decrease in sensitivity for equal doses of lower and lower energy. (See 4.8.4).

3.9.5 Rate dependence. Each radiacmeter shall retain the sensitivity, (or calibration) through as broad a range of radiation rates as possible . In any event the deviation readings taken at mid-scale from rates of 10^3, 10^5 and 10^6 R/hr the deviation between true dose and reading shall not exceed the following percentages of true dose for the range indicated.

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Radiacmeter	IM-93()/UD	.5 to 100.000	R/hr 10 percent
		100.000 to 1.5 million	R/hr 15 percent
Radiacmeter (see 4.8.5)	IM-147IO/PD	0.100	R/hr 10 percent

3.9.6 Leakage electrical. When not exposed to ionizing radiation, the leakage rate shall not exceed 0.5 percent of the full scale reading per 24 hours for all temperatures in the range of -40°F to 112°F (-40°C to + 44.4°C). The leakage rate shall not exceed 1.5 percent of the full scale reading per 24 hours at 130°F (54.4°C) and shall be scaled linearly from 0.5 percent to 1.5 percent in the temperature range of 112°F to 130°F (+44.4°C to 54.4°C) (See 4.8.6).

3.9.7 Radiacmeter IM-93()/UD. The leakage of each Radiacmeter IM-93()/UD, 8 hours after exposure to 6000 roentgens applied at exposure rates not less than 3,000 R/hr. shall not exceed 0.5 percent of full scale per 24 hours. (See 4.8.7.1).

3.9.8 Radiacmeter IM-147()/PD. The leakage of each Radiacmeter IM-147()/PS, 8 hours after exposure to 100 roengens applied at a rate of not less than 100 roentgens per hour, shall not exceed 0.5 percent of full scale per 24 hours. (See 4.8.7.2).

3.9.9 Electrostatic charger. Each radiacmeter shall be capable of being charged to zero using Radiac Charger PP-1578A. (See 4.8.8)

3.9.10 Charging contact resistance. The contact resistance of the Electrometer switch, the charging electrode, the charging end of the dosimeter and the inside surface of the charging recess shall be as low as possible. (For tests, a fully charged dosimeter shall discharge to full scale reading in less than one-half second.) (See 4.8.9).

3.9.11 Charging or zeroing voltage. Each radiacmeter shall be capable of being charged to zero with an auxiliary charger that can furnish 195 volts. (See 4.8.10)

3.9.12 Geotropism. Deflection from the normal (fiber image in the vertical position) read of the fiber image during a 360 degree rotation of each radiacmeter about its longitudinal axis while that axis is perpendicular to the force of gravity, shall not exceed ± one division on the scale. (See 4.8.11)

3.9.13 Life. After 2500 charging operations, each radiacmeter shall be capable of meeting full specification requirements for calibration accuracy (see 3 9.1). charging or zeroing voltage (see 3.9.11) and for the visual and mechanical tests. (See 4 8.12)

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3.9.14 Neutron insensitivity. The ratio of neutron to gamma dose reading shall be 0.57 ± 20%. The first collision neutron dose table for water from the National Bureau of Standard Handbook #75 shall be the dose conversion reference (see 4.8.13).

3.9.15 Field calibration. Radiacmeter IM-93()/UD shall be capable of being calibrated by Radiac Calibrator AN/UDM-2 (a SR 90 beta source). When calibrated with the AN/UDM-2, the radiacmeter shall indicate a dose of not less than 30R and preferably 300R when exposed for a period of 5 minutes. In any case, the mean value of 20 first article test samples shall be used to determine the AN/UDM-2 sensitivity for the contractor's entire production quantity. All dosimeters produced by the contractor shall have a sensitivity within ±20 percent of the established mean. (See 4.8.14)

3.9.15.1 Radiacmeter IM-147()/PD. When calibrated with AN/UDM-2 the radiacmeter shall indicate a dose of not less than 40R ± 20 percent when exposed for a period of not less than 1 min.

3.9.16 Fiber voltage polarity reversal (see 4.8.16. The fiber and radiation sensitive chamber inner wall shall be uniformly conductively coated such that a fiber initially set at midscale, shall maintain that midscale position within one minor scale division when, after setting the fiber to midscale, the charging voltage polarity is rapidly switched from positive to negative or from negative to positive.

3.9.17 Pressure. When tested in accordance with 4.8.16, the maximum allowable deviation between the reference reading before test and the test reading shall be ±5 percent.

3.10 Service conditions. The radiacmeter shall be subjected to and meet the service conditions below. For the purpose of service conditions, specified performance is defined as meeting the following requirements:

Requirement paragraph	Test Paragraph	Measurement
3.9.1	4.8.1	Calibration accuracy
3.9.3	4.8.3	Precision

3.10 1 High temperature.

a. Operating. The radiacmeter shall be operable without degradation in specified performance at ambient test temperature up to and including 130°F (+54.4°C) (See 4.9.1)

b. Storage and transportation. The radiacmeter shall withstand exposure to air temperature as high as +160°F(+71°C) (see 4.9.1).

3.10.2 Low temperature.

a. Operating. The radiacmeter shall be operable without degradation specified performance at temperature as low as -40°F(-40°C). (See 4.9.2)

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b. Storage and transportation. The radiacmeter shall withstand exposure to air temperature as low as -80°F(-62.2°C). (see 4.9.2)

3.10.3 Humidity. The radiacmeter shall exhibit no physical damage such as corrosion, rust, blistering, swelling, or deterioration of parts and materials and shall meet the specified performance when subjected to 94±4 percent relative humidity for 20 hours and 100 percent relative humidity, with condensation for 4 hours. The radiacmeter shall meet specified performance without degradation after completion of the test (see 4.9.3).

3.10.4 Altitude. The radiacmeter shall meet the specified performance at 10,000 ft (3048m) and nonoperative up to 25,000 feet. (7620m) above sea level. The radiacmeter shall meet specified performance without degradation after completion of the test (see 4.18.16).

3.10.5 Vibration. The radiacmeter shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the vibration test specified in 4.9.4.

3.10.6 Shock, transit drop. The radiacmeter shall operate without degradation in specified performance and shall suffer no mechanical damage after being subjected to the bench handling test specified in 4.9.5.

3.10.7 Salt fog. After the salt fog test of 4.9.6, the radiacmeter shall operate without degradation in specified performance and shall show no evidence of degradation such as flaking, pitting, blistering or loosening of finish on metal surface or exfoliation.

3.10.8 Temperature shock. The equipment shall be capable of withstanding temperature shocks between -40°F (-40°) and +130°F (+54.4°C), expansion or contraction of dosimeter components resulting from temperature variations shall not change the position of the fiber relative to the scale by more than ± 5 percent of full scale, (See 4.9.7)

3.11 Workmanship. The radiacmeter shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268 (see 4.11. and 4.4).

4 0 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, or purchase order the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3). Does not include packaging.

d. Inspections covered by subsidiary documents (see 4.4).

c. Quality conformance inspection of equipment before packaging (see 4.5).

4.3 First article. Unless otherwise specified in the contract, the first article inspection shall be performed by the contractor (see 3.1).

4 3.1 First article units. The contractor shall furnish 100 first article units of the radiacmeter.

4.3.2 First article inspection The first article inspection shall consist of the inspection specified in table I and shall be performed in the order specified.

4.3.3 First article data. The first article test plan and test report(s) shall be as required in the contract.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract.

Item	Where required
Parts, materials, and processes; general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.11

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TABLE I. First article inspection.

Inspections	Reqt para	Test para	Order of Tests: units 1-20	units 21-30	units 31-60	units 61-80	units 81-100
1.Inspections covered by subsidiary documents 1/	3.3 3.4 3.5 3.11	4.4	Inspection to be performed on all units				
2.Group A inspection 1/	See table II		Inspection to be performed on all units				
3.Group B inspection 1/	See table III		Inspection to be performed on all units				
4.Group C inspection							
High temperature	3.10.1	4.9.1	1				
Low temperature	3.10.2	4.9.2	2				
Humidity	3.10.3	4.9.3	4				
Altitude	3.10.4	4.8.16		1			
Vibration	3.10.5	4.9.4		2			
Shock, transit drop	3.10.6	4.9.5		3			
Salt fog	3.10.7	4.9.6		4			
Temperature Shock	3.10.8	4.9.7	3				
5.Unique first article test							
Radiacmeter IM-93	3.9.7	4.8.7.1					2
Radiacmeter IM-147	3.9.8	4.8.7.2					3
Electrostatic Charge	3.9.9	4.8.8			1		
Charging contact resistance	3.9.10	4.8.9			2		
Life	3.9.13	4.8.12			4		
Neutron insensitivity	3.9.14	4.8.13				3	
Field calibration	3.9.15	4.8.14				1	
Fiber voltage polarity reversal	3.9.16	4.8.15			3		
Energy dependence	3.9.4	4.8.4					1
Rate dependence	3.9.5	4.8.5				2	

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1/ The inspection 1 to 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements.

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4, 4.5.1 through 4.5.3.3. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspection in table II. Lots shall be formed from units that pass this inspection. Each lot shall be subjected to sampling inspection in accordance with Table III. Lots in which samples exhibit any failure shall be screened for that failure prior to being subjected to group A inspection.

TABLE II. Group A inspection.

Inspection	Requirement paragraph	Test Paragraph
Visual and mechanical		
Charging electrode diaphram	3.2.2	4.8.17
Clip	3.2.3	4.11
Cap	3.2.4	4.11
Scale	3.2.5	4.11
Radiacmeter IM-147	3.2.5.1	4.11
Radiacmeter IM-147	3.2.5.2	4.11
Materials	3.2.5.3	4.11
Optical system	3.2.6	4.12
Fiber image	3.2.6.1	4.12
Surface	3.2.7	4.11
Safety engineering	3.8	4.10
Performance requirement 1/		
Calibration accuracy	3.9.1	4.8.1
Charging or Zeroing voltage	3.9.11	4.8.10
Geotropism	3.9.12	4.8.11

1/ All performance defects are major.

4.5.1.1. Order of inspection within group A. A group A inspection shall be performed in an order satisfactory to the government.

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TABLE III. Sampling plan for Group A inspections.

Lot size	Sample size
2 to 8	1/
9 to 15	13
16 to 25	13
26 to 50	13
51 to 90	13
91 to 150	13
151 to 280	13
281 to 500	20
501 to 1200	34
1201 to 3200	42
3201 to 10,000	50
10,001 to 35,000	60
35,001 to 150,000	74
150,001 to 500,000	90
500,001 to and over	102

Notes: 1. 1/ Indicates entire lot must be inspected.
2. Acceptance number in all cases is zero.

4.5.2 Group B inspection. Group B inspection shall be performed on lots that have passed group A inspection. This inspection shall consist of the inspections listed in table IV. Lot shall be subjected to sampling inspection in accordance with table V and VI. Lots in which samples exhibit any failures shall be screened for that failure prior to the units within the lot being subject to group C inspection.

TABLE IV. Group B inspection.

Inspection	Equipment paragraph	Test paragraph
Dimensional interchangeability	3.6	4.6
Weight	3.7	4.7
Leakage	3.9.6	4.8.6
Seal	3.3.2	4.8.16
Reduced pressure	3.9.17	4.8.16

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4.5.2.1 Group B sampling plans. The group B shall be performed in an order satisfactory to the government.

TABLE V Sampling plan for interchangeability

Lot size	Sample size
2 to 8	2
16 to 15	2
16 to 25	3
26 to 50	5
51 to 90	5
91 to 150	6
151 to 280	7
281 to 500	9
501 to 1200	11
1201 to 3200	13
3201 and over	15

NOTE: Acceptance number in all cases is zero.

TABLE VI. Sampling plan for weight

Lot size	Sample size
2 to 8	3
9 to 15	3
16 to 25	3
26 to 50	5
51 to 90	6
91 to 150	7
151 to 280	10
281 to 500	11
1201 to 3200	18
3201 to 10,000	22
10,000 and over	29

NOTE: Acceptance number in all cases is zero.

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed group A and group B inspections. The inspection shall consist of the inspections specified in table VII. Samples shall be selected in accordance with 4.5 3.1.

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4.5.3.1 Sampling for group C inspection. Forty units shall be selected at ...om, from the first 400 production models, ten samples for each subgroup in ...le VII. Those units shall constitute the group C requirement for the first 400 units produced. Thereafter, 8 units shall be selected at random, for each subgroups 1, 2, 3 and 4 in table VII, from every 400 units or fraction therefore produced. If the quantity of the last fraction produced consist of less than 200 units, no group C inspection shall be required. The sample units shall be selected from the conformance lots consisting of units from each 400 units or fraction thereof produced. The salt fog, life inspections shall be performed at the mid-point of the production contract.

4.5.3.2 Group C failures. Action required relative to group C failure shall be as specified in the contract.

4.5.3.3 Reinspection of conforming group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass group A inspection, with the exception of safety engineering.

TABLE VII. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1		
High temperature	3.10.1	4.9.1
Low temperature	3.10.1	4.9.2
Temperature shock	3.10.8	4.9.8
Humidity	3.10.3	4.9.3
Subgroup 2		
Altitude	4.10.4	4.9.4
Vibration	4.10.5	4.9.5
Shock, transit drop	3.10.6	4.9.6
Salt fog 1/	3.10.7	4.9.7
Subgroup 3		
Linearity	3.9.1	4.8.1
Precision	3.9.3	4.8.3
Life	3.9.13	4.8.12
Field Calibration	3.9.15	4.8.14
Subgroup 4.2/		
Energy dependence	3.9.4	4.8.4
Rate dependence	3.9.5	4.8.5
Neutron insensitivity	3.9.14	4.8.13

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1/ The equipment shall be thoroughly washed, cleaned, dried and refurbished after this inspection before proceeding with subsequent inspection.

2/ Subgroup 4 testing shall be done only 3 times during production at 1/3, 2/3 and end of production.

4.6 Interchangeability. The dimensions of radiacmeters shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.6. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

Item	Gage list
Radiacmeter IM-93()/UD	SC-GL-57913
Radiacmeter IM-147()/PD	SC-GL-57914

4.7 Weight. The radiacmeters shall be weighed. The total weight shall not exceed that given in 3.7.

4.8 Performance tests.

4.8.1 Calibration accuracy. Calibration shall be checked by the use of Cobalt 60. Free-space conditions as possible will be realized for this calibration. The Government reserves the right to cross check calibration at a US Government facility,. The Government test shall govern in the event of conflict between the contractor's results and the Government's results. Calibration accuracy shall be determined at 1/4, 1/2. 3/4 and full scale and shall satisfy the requirement of 3.9.1. The true dose rate shall be determined by a Victoreen R-meter or equal that has been calibrated by the US National Bureau of Standards for a US contractor or the National Primary Calibration Agency of the respective nation for a foreign contractor. Calibration of Radiacmeter IM-93()/UD shall be performed at any dose rate between 100 R/hr and 500 R/hr. Calibration of the Radiacmeter IM-147() /PD shall be performed at an dose rate of 50 R/hr and 100 R/hr.

4.8.2 Linearity test. The radiacmeters shall be tested for compliance with 3.9.2.

4.8.3 Precision test. The radiacmeters shall be tested for compliance with 3.9.3. Each radiacmeter shall be identically exposed for 5 exposures. All readings shall fall within ±2 1/2% of the average reading for the particular instrument.

4.8.4 Energy dependence calibration tests. The air equivalence, of the radiacmeters over the energy range of 30 KeV to 5.0 MeV shall be tested. The highest test energy required shall be satisfied using C060. (See 3.9.4)

4.8.4 1 Test equipment.

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4.8.4.1.1 X-Ray Potential. A constant potential X-ray machine operable up to maximum voltage of 250 KV shall be employed for calibration tests in the low energy region.

4.8.4.1.2 Calibrating system. A diaphraming or collimating system shall be used to define the X-ray beam so that the instrument being tested will be irradiated uniformly and scattering from walls or neighboring objects will be minimized.

4.8.4.1.3 Effective wavelength. The data on X-ray tube potentials, filtration, energy at maximum intensity and the half height limits on energy spread as obtained by Krammer's rule are shown in Table VIII. A Westinghouse quadrocandex 250 KVP constant potential machine or equal associated with an X-ray tube with an inherent filtration equal to 3 millimeter of aluminum is required.

Table VIII. X-ray tube data

Tube Applied Potential Kilovolts	Filtration in Millimeters	Beam Energy at Maximum Intensity Kilo-electron volts	Energy Spread at Half Heights Kilo-electron volts
40	1.67 Al	30	-5.3, +8.0
70	0.89 Cu	56	-10.1, +10.05
100	1.07 Sn	85	-14.0, +10.5
150	4.28 Sn	129	-19.0, +16.3
	2.00 Pb	178	-23.0, +15.5
200	2.14 Sn		
250	4.00 Pb	222	-28.0, +21.5
	2.14 Sn		

4.8.4.2 Test procedure. Expose the instrument to various dose rates over the specified range of the equipment using a Cobalt 60 source and record the readings. (see 4.8.1) The instrument shall than be exposed to the same true dose rates as above, however, these rates shall have various energies over the range of 80 KeV to 5.0 MeV. The true dose rate shall be determined by a Victoreen R-meter, or equal, that has been calibrated by the National Bureau of Standards. Record the instrument's reading for each exposure. These values shall than be compared to those obtained from Cobalt 60 to determine compliance with paragraph 3.9.4. The instrument shall then be exposed to the same true dose rates as above but the energies involved shall be below 80 KeV and the results obtained shall comply with paragraph 3.9.4.

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4.8.5 Rate dependence taste. Each radiacmeter shall be tested to determine compliance with the requirements of 3.9,5. A 2 MeV Van de Graff generator,may be used in testing. Calibrated thermoluminescent dosimeter (TLD.) may be used to determine total exposure.

4.8.6 Leakage (electrical). After continuous exposure totaling 2000 R given at exposure rates between 1,000 and 10,000 R/hr, the dosimeter shall be set to zero. Electrical leakage in a continuous 4 hour period, starting no more than 10 minutes after exposure is completed, shall not exceed 5 percent of full scale. Beginning 48 hours after the exposure is completed, the electrical leakage shall not exceed 3% of full scale upon completion of a continuous test period of an additional 144 hours. The above test to be performed at 112°F±2°F temperature. Upon completion of the above test, charge the radiacmeter to zero at room temperature. The electrical leakage shall not exceed 9% of full scale as a result of exposure to a temperature of 130°F ±5%°F for a test period of 144 hours commencing 8 hours after temperature stabilization has been reached. The instruments shall be read at room temperature. (See 3.9.6)

4.8.7 Leakage after exposure to high radiation tests.

4.8.7.1 Radiacmeter IM-93()/UD. The Radiacmeter IM-93()/UD shall be subjected to a dose of 6000 roentgens as specified in 3.9.7. Eight hours after the start of exposure the instrument shall be charged to 200 roentgens and retained for 24 hours At the end of this time, the units shall comply with 3.9.7

4.8.7.2 Radiacmeter IM-147()/PD. The Radiacmeter IM 147()PD shall be subjected to a dose of 100 roentgens as specified in 3.9.8. Eight hours after the start of exposure, the instrument shall be charged to 25 roentgens and retained for 24 hours. At the end of this time, the units shall comply with 3.9.8.

4.8.8 Electrostatic charger. The Radiacmeter shall be depressed on the charging pedestal of Radiac Charger PP-1578A and charged to zero Units shall comply with the requirements of 3.9.9 and 3.9.10.

4.8.9 Charging contact resistance. The dosimeter shall be charged to zero. The charging electrode shall be depressed and shall be short circuited to the inside of the charging recess. The dosimeter shall comply with the requirements of 3.9.10.

4.8.10 Charging or zeroing voltage test. The dosimeter shall be charged with an auxiliary charger to determine compliance with 3.9.11.

4.8.11 Geotropism. The equipment shall be tested to determine compliance with 3 9.12.

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4.8.12. Life. The equipment shall be tested to determine compliance with 3.9.13. The instrument will be depressed on a conventional spring loaded charging socket by means of a cam driven reciprocating drive operating at a rate of approximately 900 strokes per hour. Charging potential will be applied to the socket in such a manner that the internal system will undergo electrical charge fluctuations of the same time that mechanical stress are being applied to the diaphragm assembly. After 2500 charging operations, the instrument shall be tested for calibration accuracy (see 4.8.1), charging or zeroing voltage (see 4.8.9, and for the visual and mechanical test (see 4.8.17).

4.8.13 Neutron insensitivity (see 3.9.14). Each type of radiacmeter shall be tested for insensitivity to fission spectrum neutrons as follows:

a. Charge the radiacmeter to zero.

b. Expose the radiacmeter to 100 rods of fission spectrum neutrons or equivalent. Observe and record the scale reading.

c. Substract calibrated gamma dose to obtain neutron dose reading.

d. Recharge to zero.

e. Expose radiacmeter to 100 rods of gamma radiation (Cobalt 60). Observe and record the scale reading.

f. The ratio of neutron to gamma reading shall not exceed 0.57 $\pm$ 20%.

g. The ratio of neutron to gamma readings will be established in preproduction testing.

NOTE: Paragraph 4.8.13 defines a method of test to assure compliance with the neutron sensitivity requirements of paragraph 3.9.14. Method of testing is dependent upon the test facilities available to the contractor therefore the contractor shall specify in his test procedure how testing will be done to meet the requirements of paragraph 3.9.14.

4.8.14 Field calibration test. When calibrated in Radiac Calibrator AN/UDM-2 in accordance with the operating instructions provided with the test set, Radiacmeter IM-93()/UD shall respond in accordance with the requirements of 3.9.15 of this specification.

4.8.15 Fiber voltage polarity reversal. The Radiacmeter shall be tested as follows:

a. The Radiacmeter shall be placed on the charging pedestal of a charger modified to switch the polarity of the charging voltage to the pedestal.

b. The Radiacmeter fiber shall be set to a midscale reading.

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c. With the Radiacmeter still in contact with the charging pedestal, and the fiber at midscale, the polarity of the charging voltage shall be reversed.

d. The instrument shall comply with the requirements of 3.4.16.

4.8.16 Reduced pressure test. Each type of radiacmeter, using bell jar and vacuum pump techniques, shall be subjected to a reduced pressure lower than 0.1 mm of mercury for a continuous period of 72 hours. The radiacmeter shall be simultaneously, exposed to gamma radiation and the radiation sensitivity measurement compared to that obtained before this test. No loss of sensitivity shall be greater than ±5% between reference reading, before and after test. This test shall be performed on samples subjected to and passed the Precision test paragraph 3.9.3. (See 3.9.17)

4.8.17 Charging electrode diaphragm. The force required for the charging electrode to make electrical contact shall be gaged. The axial and radial forces that the electrode shall be capable of withstanding shall be tested to determine compliance with 3.2.2.

4.9 Service conditions. The service conditions tests shall be performed as detailed below. Each radiacmeter subjected to these tests shall have passed group A and B tests and shall be resubmitted to and pass the group A tests with the exception of safety engineering inspection after completion of all service condition tests. Where operation of the test item is required during any service condition test, the test item shall be required to pass the specified performance and additional requirements when noted. (See 3.10)

4.9.1 High temperature. The radiacmeter shall be subjected to the test of Method 501.1, Procedure II of MIL-STD-810. The storage temperature (Step 4) shall be 160°F (71°C). In Step 7, the operating temperature shall be +130°F (+54.4°C) (See 3.10.1).

4.9.2 Low temperature. The radiacmeter shall be subjected to the test of Method 502.1, Procedure I of MIL-STD-810. The storage temperature (Step 2) shall be -80°F (-62°C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (Step 4) shall be -40°F (-40°C) (see 3 10.2).

4.9.3 Humidity. The radiacmeter shall be subjected to the test of Method 507.1, Procedure III of MIL-STD-810. In Step 6, measurements shall be taken during the last 5 hours of the last cycle (see 3.10.3).

4.9.4 Vibration. The radiacmeter shall be tested as follows:

a. The radiacmeter shall be fastened on a vibration table that can be controlled within 10 percent of the specified vibration.

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b. The radiacmeter shall be vibrated with the frequency varying between 10 and 55 cycles per second at a total excursion of 1/64 of an inch, the frequency varying uniformly from 10 to 55 cycles per second and returning to 10 cycles per second in approximately one minute.

c. The vibration specified above shall be applied in each of three mutually perpendicular planes; one of which shall be along the longitudinal axis of the instrument. The vibration shall be applied for a period of 30 minutes in each of the three planes.

d. Following this test, the instrument shall be tested for calibration accuracy (see 4.8.2) leakage (see 4.8.6), charging or zeroing voltage (see 4.8.10), reduced pressure (see 4.8.16) and for visual and mechanical defects (see 4.5.1).

e. The equipment shall comply with requirements of 3.10.5.

Note: Leakage test shall be of 3 day duration in lieu of 6 days

4.9.5 Shock, drop. The equipment shall be dropped a distance of four feet, two drops on each end and two drops on the side, to a rigid wooden surface. Following this test, the instrument shall be tested for calibration accuracy (see 4.8.2), leakage (see 4.8.6), charging or zeroing voltage (see 4.8.10),
and reduced pressure (see 4.8.16). This test may be preformed under any operating environmental condition specified under paragraph 3.10.

Note: Leakage test shall be of 3 day duration in lieu of 6 days.

4.9.6 Salt fog. The radiacmeter shall be subjected to the test of Method 509 1, Procedure I of MIL-STD-810 (see 3.10.7).

4.9.7 Thermo-mechanical equilibrium. The instruments shall be zeroed at room temperature and then placed at 130°F for sufficient time to insure a condition of thermal equilibrium (approximately 2 hours) at which time they shall be read, and then returned to room temperature for 2 hours and read again. Deflections which may occur will be considered due to mechanical changes if they are in a direction opposite to that due to electrical charge leakage. Thus, a down scale deflection at 130°F accompanied by an equivalent up scale deflection when the instrument is returned to room temperature is a mechanical shift. If an up scale deflection occurs partially or not at all, this may be considered in part an electrical charge leakage. This test shall be similarly applied at 40°F. The equipment shall meet the requirement of 3.10.8.

4.10 Safety engineering inspection. A visual inspection shall be performed in the presence of a government safety representative to determine that all requirements of 3.8 have been incorporated in the equipment. Inability to meet these requirements shall constitute a failure of the test. Results of this inspection shall be included in the Safety Assessment Report, if required in the contract.

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4.11 Visual and mechanical inspection. These inspections shall be performed in any order which is satisfactory to the government. The units shall be examined for the applicable defects listed in MIL-STD-252 (see 3.8 and 3.11).

4.12 Readability. The radiacmeter shall be exposed to an illumination of 0.5 foot candles to determine compliance with 3.2.6 and 3.2.6.1.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity.

6. NOTES (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Radiacmeters IM-93()/UD and IM-147()/PD are hermetically sealed, ionization chamber-quartz fiber type dosimeter capable of detecting and measuring X-rays and gamma radiation. Radiacmeter IM-93()/UD has a range of 0 to 600 Roentgens. Radiacmeter IM-147()/UD which replaces Radiacmeter IM-20()/UD, has a range of 0 to 50 Roentgens. Both are portable instruments indicate accumulated amount of radiation to which the bearer has been exposed.

6.2 Ordering data. Procurement documents should specify the following:

a Title, number and date of this specification and any amendment thereto.

b. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

c. Marking and shipping of samples.

d. Place of final inspection.

e. Number of first articles to be submitted for approval (see 3.1).

6.3 Subject terms (keywords) list.

Radiacmeter IM-93()/UD
Radiacmeter IM-147()/PD
Measurements of X-rays and gamma radiation
Nuclear environments.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-CR

Preparing activity:
Army - CR
Project 6665-A637

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I RECOMMEND A CHANGE:	1 DOCUMENT NUMBER MIL-R-13029C(CR)	2 DOCUMENT DATE (YYMMDD) 13 May 1992

3 DOCUMENT TITLE RADIACMETER IM-93()/UD, RADIACMETER IM-147()/PD

4 NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed)*

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a NAME U.S. Army Communications-Electronics Command and Fort Monmouth	b TELEPHONE (Include Area Code) (1) Commercial 908-532-5851 (2) AUTOVON 992-5851
c ADDRESS (Include Zip Code) ATTN: AMSEL-ED-TM Fort Monmouth, NJ 07703-5023	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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